

Unit 5. Systems of Equations and Elimination

1 Elementary Operations and Echelon Forms

1.1 Elementary matrices and row equivalence

Definition 5.1 (Elementary row operations and matrices). For $i \neq j$ and $\lambda \neq 0$, the elementary row operations on an $m \times n$ matrix are

$$R_i \leftrightarrow R_j, \quad R_i \leftarrow \lambda R_i, \quad R_i \leftarrow R_i + \lambda R_j.$$

The corresponding elementary matrices are obtained by applying the operation to $\mathbf{I}_m$:

$$\begin{aligned}\mathbf{E}_{ij} &= \mathbf{I}_m - (\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)', \\ \mathbf{E}_i(\lambda) &= \mathbf{I}_m + (\lambda - 1)\mathbf{e}_i\mathbf{e}_i', \\ \mathbf{E}_i(\lambda \mid j) &= \mathbf{I}_m + \lambda\mathbf{e}_i\mathbf{e}_j' .\end{aligned}$$

The elementary column operations $C_i \leftrightarrow C_j$, $C_i \leftarrow \lambda C_i$, and $C_i \leftarrow C_i + \lambda C_j$ are represented by postmultiplication by

$$\mathbf{E}_{ij}, \quad \mathbf{E}_i(\lambda), \quad \mathbf{E}_j(\lambda \mid i),$$

respectively, with matrices of order n .

Origin: A&M §6.1, Ex. 6.3, and Ex. 6.10

Theorem 5.2 (Elementary action and inverse). Premultiplication by $\mathbf{E}_{ij}$, $\mathbf{E}_i(\lambda)$, or $\mathbf{E}_i(\lambda \mid j)$ performs its defining row operation; the three postmultiplications in Definition 5.1 perform the defining column operations. Each elementary matrix is nonsingular, with

$$\mathbf{E}_{ij}^{-1} = \mathbf{E}_{ij}, \quad \mathbf{E}_i(\lambda)^{-1} = \mathbf{E}_i(\lambda^{-1}), \quad \mathbf{E}_i(\lambda \mid j)^{-1} = \mathbf{E}_i(-\lambda \mid j).$$

Their determinants are -1 , λ , and 1 , respectively.

Origin: A&M Ex. 6.2, 6.5, and 6.9

Proof. For every column $\mathbf{a}$, the three products replace its coordinates by

$$(a_1, \dots, a_j, \dots, a_i, \dots, a_m)', \quad (a_1, \dots, \lambda a_i, \dots, a_m)',$$

and

$$(a_1, \dots, a_i + \lambda a_j, \dots, a_m)',$$

with the first expression interchanging coordinates i, j . Applying the stated reverse operation gives $\mathbf{a}$, hence gives $\mathbf{I}_m$ on every column. The column claims are the transposes of the row claims. The determinants follow from the permutation formula, diagonal multiplication, and linearity of the determinant in row i , respectively:

$$\det \mathbf{E}_{ij} = -1, \quad \det \mathbf{E}_i(\lambda) = \lambda, \quad \det \mathbf{E}_i(\lambda \mid j) = 1 + \lambda \det(\mathbf{I}_m \text{ with } R_i = R_j) = 1.$$

□

Definition 5.3 (Row equivalence). Matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ are **row equivalent**, written $\mathbf{A} \sim_r \mathbf{B}$, if

$$\mathbf{B} = \mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A}$$

for elementary $m \times m$ matrices $\mathbf{E}_1, \dots, \mathbf{E}_k$; the empty product is $\mathbf{I}_m$.

Origin: A&M §6.1 and Ex. 6.19

Theorem 5.4 (Invariants of row equivalence). If $\mathbf{B} = \mathbf{E}\mathbf{A}$, where $\mathbf{E}$ is a product of elementary matrices, then

$$\text{row}(\mathbf{B}) = \text{row}(\mathbf{A}), \quad \text{null}(\mathbf{B}) = \text{null}(\mathbf{A}), \quad \text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A}), \quad \text{col}(\mathbf{B}) = \mathbf{E} \text{col}(\mathbf{A}).$$

Thus row equivalence is an equivalence relation. It need not preserve the column space as a subset of $\mathbb{R}^m$.

Origin: Course formulation from A&M Ex. 6.2, 6.10, and 6.19

Proof. Theorem 5.2 makes $\mathbf{E}$ nonsingular. Hence

$$\text{row}(\mathbf{E}\mathbf{A}) \subseteq \text{row}(\mathbf{A}), \quad \mathbf{A} = \mathbf{E}^{-1}\mathbf{B} \implies \text{row}(\mathbf{A}) \subseteq \text{row}(\mathbf{B}),$$

and

$$\mathbf{B}\mathbf{x} = \mathbf{0} \iff \mathbf{E}\mathbf{A}\mathbf{x} = \mathbf{0} \iff \mathbf{A}\mathbf{x} = \mathbf{0}.$$

Equality of row spaces gives equality of ranks. Also

$$\text{col}(\mathbf{B}) = \{\mathbf{E}\mathbf{y} : \mathbf{y} \in \text{col}(\mathbf{A})\} = \mathbf{E} \text{col}(\mathbf{A}).$$

Reflexivity uses $\mathbf{I}_m$, symmetry uses $\mathbf{E}^{-1}$, and transitivity uses products. □

1.2 Echelon forms and pivots

Definition 5.5 (REF, RREF, pivots, and free columns). A matrix is in **row-echelon form** (REF) if

- (i) every zero row follows every nonzero row;
- (ii) the first nonzero entry of each nonzero row is 1, called its **pivot**;
- (iii) successive pivot columns have strictly increasing indices, and all entries below a pivot are zero.

It is in **reduced row-echelon form** (RREF) if every pivot is the only nonzero entry in its column. A pivot column contains a pivot; every other column is **free**.

Origin: Course formulation from A&M §6.2 and Ex. 6.13

Theorem 5.6 (Existence of echelon forms). For every $\mathbf{A} \in \mathbb{R}^{m \times n}$, products of elementary matrices $\mathbf{E}$ and $\tilde{\mathbf{E}}$ exist such that

$$\mathbf{H} = \mathbf{E}\mathbf{A} \quad \text{is in REF}, \quad \mathbf{R} = \tilde{\mathbf{E}}\mathbf{A} \quad \text{is in RREF}.$$

Origin: A&M Ex. 6.13 and 6.16

Proof. If $\mathbf{A} = \mathbf{O}$, take $\mathbf{E} = \mathbf{I}_m$. Otherwise let j be the least index of a nonzero column. A row interchange places a nonzero entry in position $(1, j)$; scaling row 1 makes it 1; row additions clear the entries below it. The remaining active submatrix has rows $2, \dots, m$ and columns $j+1, \dots, n$. Induction on $\min\{m, n\}$ gives REF. Starting with the last pivot and proceeding upward, row additions clear the entries above each pivot. The resulting matrix is in RREF. Every operation is represented by an elementary matrix, so their finite products give $\mathbf{E}$ and $\tilde{\mathbf{E}}$. $\square$

Theorem 5.7 (Pivot data). Let $\mathbf{R} = \mathbf{EA}$ be an RREF of $\mathbf{A} \in \mathbb{R}^{m \times n}$, with pivot columns $p_1 < \dots < p_r$ and free columns $q_1 < \dots < q_{n-r}$. Then

$$\text{rank}(\mathbf{A}) = r, \quad \{\mathbf{a}_{p_1}, \dots, \mathbf{a}_{p_r}\} \text{ is a basis of } \text{col}(\mathbf{A}).$$

For $\mathbf{t} = (t_1, \dots, t_{n-r})'$, all members of $\text{null}(\mathbf{A})$ are given by

$$x_{q_j} = t_j, \quad x_{p_i} = - \sum_{j=1}^{n-r} r_{iq_j} t_j.$$

Setting $\mathbf{t} = \mathbf{e}_j$ gives a basis of $\text{null}(\mathbf{A})$. For $r = n$, this basis is empty and $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$.

Origin: Course formulation from A&M Ex. 6.12, 6.20, and 6.36

Proof. The r nonzero rows of $\mathbf{R}$ are independent: in a zero linear combination, inspection of pivot columns $p_1, \dots, p_r$ gives each coefficient equal to zero. The remaining rows are zero; hence $\text{rank}(\mathbf{R}) = r = \text{rank}(\mathbf{A})$.

The pivot columns of $\mathbf{R}$ are $\mathbf{e}_1, \dots, \mathbf{e}_r$. Therefore

$$\sum_{i=1}^r c_i \mathbf{a}_{p_i} = \mathbf{0} \iff \sum_{i=1}^r c_i \mathbf{e}_i = \mathbf{0} \iff c_1 = \dots = c_r = 0.$$

For every free column,

$$\mathbf{E} \mathbf{a}_{q_j} = \sum_{i=1}^r r_{iq_j} \mathbf{e}_i = \mathbf{E} \sum_{i=1}^r r_{iq_j} \mathbf{a}_{p_i};$$

nonsingularity of $\mathbf{E}$ gives spanning.

Finally, $\mathbf{A} \mathbf{x} = \mathbf{0} \iff \mathbf{R} \mathbf{x} = \mathbf{0}$, and the i th nonzero row of $\mathbf{R} \mathbf{x} = \mathbf{0}$ is

$$x_{p_i} + \sum_{j=1}^{n-r} r_{iq_j} x_{q_j} = 0.$$

The displayed parametrisation follows. Its vectors for $\mathbf{t} = \mathbf{e}_j$ span the null space and are independent because their free coordinates are the standard basis. $\square$

Example 5.1 (One reduction, three spaces). For

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & 1 & 3 \\ 0 & 0 & 3 & 3 \end{pmatrix} \sim_r \mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

the pivot columns are 1, 3. Hence

$$\text{rank}(\mathbf{A}) = 2, \quad \text{col}(\mathbf{A}) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right\},$$

and

$$\text{null}(\mathbf{A}) = \left\{ s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

Origin: Course formulation from A&M Ex. 6.14 and 6.20

1.3 Normal form under row and column operations

Theorem 5.8 (Rank normal form). If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has rank r , then products of elementary matrices $\mathbf{E} \in \mathbb{R}^{m \times m}$ and $\mathbf{F} \in \mathbb{R}^{n \times n}$ satisfy

$$\mathbf{EAF} = \mathbf{J}_r := \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

Consequently, two $m \times n$ matrices can be transformed into each other by elementary row and column operations if and only if they have the same rank.

Origin: A&M Ex. 6.16–6.17 and 6.19

Proof. For $r = 0$, $\mathbf{A} = \mathbf{O} = \mathbf{J}_0$, so take identity factors. Assume $r > 0$. Reduce $\mathbf{A}$ to an RREF $\mathbf{R}$. Column interchanges place its pivot columns first, giving

$$\begin{pmatrix} \mathbf{I}_r & \mathbf{G} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

For each column of $\mathbf{G}$, subtract its linear combination of the first r columns. The resulting column-operation product gives $\mathbf{J}_r$.

Elementary operations preserve rank. Conversely, if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}) = r$, choose

$$\mathbf{E}_A \mathbf{A} \mathbf{F}_A = \mathbf{J}_r = \mathbf{E}_B \mathbf{B} \mathbf{F}_B.$$

Then

$$\mathbf{B} = \mathbf{E}_B^{-1} \mathbf{E}_A \mathbf{A} \mathbf{F}_A \mathbf{F}_B^{-1},$$

and every inverse factor is elementary. □

Corollary 5.9 (Elementary factorisation of a nonsingular matrix). A square matrix is nonsingular if and only if it is a finite product of elementary matrices.

Origin: A&M Ex. 6.18

Proof. If $\mathbf{A}$ is nonsingular, Theorem 5.8 gives $\mathbf{E}\mathbf{A}\mathbf{F} = \mathbf{I}_n$, so

$$\mathbf{A} = \mathbf{E}^{-1}\mathbf{F}^{-1},$$

a product of elementary matrices. The converse follows because every elementary matrix is nonsingular and a product of nonsingular matrices is nonsingular. $\square$

2 Linear Systems

2.1 Consistency and complete solution sets

Definition 5.10 (Linear system and consistency). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times p}$,

$$\mathbf{A}\mathbf{X} = \mathbf{B}$$

is a **linear system** in $\mathbf{X} \in \mathbb{R}^{n \times p}$. The matrices $\mathbf{A}$, $\mathbf{B}$, and $(\mathbf{A} : \mathbf{B})$ are its **coefficient matrix**, **right-hand side**, and **augmented matrix**. A value of $\mathbf{X}$ satisfying the equality is a **solution**. The system is **consistent** if a solution exists and **homogeneous** if $\mathbf{B} = \mathbf{O}$.

Origin: H §§7.1–7.2; A&M §§6.4–6.5

Theorem 5.11 (Classification of a linear system). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times p}$, the following are equivalent:

$$\mathbf{A}\mathbf{X} = \mathbf{B} \text{ is consistent,} \quad \text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A}), \quad \text{rank}(\mathbf{A} : \mathbf{B}) = \text{rank}(\mathbf{A}).$$

If $\mathbf{X}_0$ is one solution, then the complete solution set is

$$\{\mathbf{X} : \mathbf{A}\mathbf{X} = \mathbf{B}\} = \{\mathbf{X}_0 + \mathbf{Z} : \mathbf{A}\mathbf{Z} = \mathbf{O}\}.$$

It contains exactly one member if and only if $\text{rank}(\mathbf{A}) = n$. For $p = 1$, $\dim \text{null}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$.

Origin: H Thm. 7.2.1 and Lemma 11.3.1; A&M Ex. 6.42, 6.44, and 6.45

Proof. Writing $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_p)$ and $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_p)$,

$$\mathbf{A}\mathbf{X} = \mathbf{B} \iff \mathbf{b}_j = \mathbf{A}\mathbf{x}_j \in \text{col}(\mathbf{A}) \quad (1 \leq j \leq p) \iff \text{col}(\mathbf{A} : \mathbf{B}) = \text{col}(\mathbf{A}).$$

Taking dimensions gives the rank condition. If $\mathbf{A}\mathbf{X}_0 = \mathbf{B}$, then

$$\mathbf{A}\mathbf{X} = \mathbf{B} \iff \mathbf{A}(\mathbf{X} - \mathbf{X}_0) = \mathbf{O}.$$

Uniqueness is equivalent to $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$, hence to $\text{rank}(\mathbf{A}) = n$ by rank-nullity. The final dimension is rank-nullity. $\square$

Corollary 5.12 (Square systems). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the following are equivalent:

$$\mathbf{Ax} = \mathbf{b} \text{ has a unique solution for every } \mathbf{b} \in \mathbb{R}^n,$$

$$\text{rank}(\mathbf{A}) = n, \quad \text{null}(\mathbf{A}) = \{\mathbf{0}\}, \quad \det(\mathbf{A}) \neq 0, \quad \mathbf{A} \text{ is nonsingular.}$$

In this case the solution is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Origin: H §11.1; A&M Ex. 6.34 and 6.44; S §11.1

Proof. Theorem 5.11 gives the equivalence of the first three conditions. The inverse, determinant, and rank criteria for nonsingularity give the remaining equivalences and the displayed solution. $\square$

2.2 Compatibility and row relations

Definition 5.13 (Compatibility). The system $\mathbf{AX} = \mathbf{B}$ is **compatible** if

$$\mathbf{k}'\mathbf{A} = \mathbf{0}' \implies \mathbf{k}'\mathbf{B} = \mathbf{0}' \quad (\mathbf{k} \in \mathbb{R}^m).$$

Thus every linear relation among the rows of $\mathbf{A}$ must hold among the corresponding rows of $\mathbf{B}$.

Origin: H §7.3 preceding Thm. 7.3.1

Theorem 5.14 (Consistency equals compatibility). The system $\mathbf{AX} = \mathbf{B}$ is consistent if and only if it is compatible.

Origin: H Thm. 7.3.1

Proof. If $\mathbf{AX}_0 = \mathbf{B}$, then

$$\mathbf{k}'\mathbf{A} = \mathbf{0}' \implies \mathbf{k}'\mathbf{B} = \mathbf{k}'\mathbf{AX}_0 = \mathbf{0}'.$$

Conversely, let rows $\mathbf{a}'_{i_1}, \dots, \mathbf{a}'_{i_r}$ form a basis of $\text{row}(\mathbf{A})$, and let $\mathbf{A}_0, \mathbf{B}_0$ contain the corresponding rows of $\mathbf{A}, \mathbf{B}$. If $r > 0$, then $\mathbf{A}_0$ has full row rank, so $\mathbf{A}_0\mathbf{X}_0 = \mathbf{B}_0$ has a solution. For every remaining row, write

$$\mathbf{a}'_i = \sum_{j=1}^r \lambda_{ij} \mathbf{a}'_{i_j}.$$

Compatibility applied to this row relation gives

$$\mathbf{b}'_i = \sum_{j=1}^r \lambda_{ij} \mathbf{b}'_{i_j}, \quad \mathbf{a}'_i \mathbf{X}_0 = \sum_{j=1}^r \lambda_{ij} \mathbf{a}'_{i_j} \mathbf{X}_0 = \mathbf{b}'_i.$$

Hence $\mathbf{A}\mathbf{X}_0 = \mathbf{B}$. If $r = 0$, then $\mathbf{A} = \mathbf{O}$; compatibility with $\mathbf{k} = \mathbf{e}_i$ gives $\mathbf{B} = \mathbf{O}$, and $\mathbf{X}_0 = \mathbf{O}$ is a solution. $\square$

Example 5.2 (A compatibility certificate). For

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 3 & 1 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{k}'\mathbf{A} = \mathbf{0}',$$

the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ can be consistent only if

$$\mathbf{k}'\mathbf{b} = -b_1 - b_2 + b_3 = 0.$$

For $\mathbf{b} = (1, 2, 4)'$, this scalar equals 1, so the system is inconsistent. For $\mathbf{b} = (1, 2, 3)'$, row reduction gives $\mathbf{x} = (1, 0)'$.

Origin: Course formulation from H Thm. 7.3.1

2.3 Augmented elimination

Theorem 5.15 (Tracking the augmented matrix). Let $\mathbf{b} \in \mathbb{R}^m$, and let $\mathbf{E}$ be nonsingular. If

$$(\mathbf{R} : \mathbf{c}) = \mathbf{E}(\mathbf{A} : \mathbf{b}),$$

then

$$\mathbf{Ax} = \mathbf{b} \iff \mathbf{Rx} = \mathbf{c}.$$

If $(\mathbf{R} : \mathbf{c})$ is in REF, the system is inconsistent exactly when its last column is a pivot column. Otherwise the pivot variables are determined by the free variables; their number is $n - \text{rank}(\mathbf{A})$.

Origin: Course formulation from A&M Ex. 6.41–6.47

Proof. Nonsingularity of $\mathbf{E}$ gives

$$\mathbf{Ax} = \mathbf{b} \iff \mathbf{EAx} = \mathbf{Eb} \iff \mathbf{Rx} = \mathbf{c}.$$

A pivot in the last column produces the equation $0 = 1$. Without such a pivot, every nonzero row has a coefficient pivot; solving the pivot equations from the last upward expresses each pivot variable in terms of the free variables. The pivot count is $\text{rank}(\mathbf{A})$, by Theorem 5.7. $\square$

Example 5.3 (Exact augmented reduction).

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \end{array} \right) \sim_r \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \sim_r \left(\begin{array}{ccc|c} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right).$$

With $x_2 = t$,

$$\mathbf{x} = \begin{pmatrix} 2 - 2t \\ t \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Origin: Course formulation from A&M Ex. 6.41 and 6.46

3 Triangular Solution and Factorisation

3.1 Forward and back substitution

Theorem 5.16 (Triangular substitution). Let $\mathbf{U} = (u_{ij}) \in \mathbb{R}^{n \times n}$ be upper triangular with $u_{ii} \neq 0$. The unique solution of $\mathbf{U}\mathbf{x} = \mathbf{b}$ is generated by

$$x_n = \frac{b_n}{u_{nn}}, \quad x_i = \frac{b_i - \sum_{j=i+1}^n u_{ij}x_j}{u_{ii}} \quad (i = n-1, \dots, 1).$$

Let $\mathbf{L} = (\ell_{ij}) \in \mathbb{R}^{n \times n}$ be lower triangular with $\ell_{ii} \neq 0$. The unique solution of $\mathbf{L}\mathbf{x} = \mathbf{b}$ is generated by

$$x_1 = \frac{b_1}{\ell_{11}}, \quad x_i = \frac{b_i - \sum_{j=1}^{i-1} \ell_{ij}x_j}{\ell_{ii}} \quad (i = 2, \dots, n).$$

For nonsingular upper block-triangular $\mathbf{A} = (\mathbf{A}_{ij})_{i,j=1}^s$, with conformable partitions $\mathbf{X} =$

$(\mathbf{X}'_1, \dots, \mathbf{X}'_s)'$ and $\mathbf{B} = (\mathbf{B}'_1, \dots, \mathbf{B}'_s)'$,

$$\mathbf{X}_i = \mathbf{A}_{ii}^{-1} \left(\mathbf{B}_i - \sum_{j=i+1}^s \mathbf{A}_{ij} \mathbf{X}_j \right) \quad (i = s, \dots, 1).$$

The lower block-triangular formula is

$$\mathbf{X}_i = \mathbf{A}_{ii}^{-1} \left(\mathbf{B}_i - \sum_{j=1}^{i-1} \mathbf{A}_{ij} \mathbf{X}_j \right) \quad (i = 1, \dots, s).$$

The upper recursions are **back substitution**; the lower recursions are **forward substitution**.

Origin: H Eqs. 11.8.2–11.8.6

Proof. The i th upper-triangular equation is

$$u_{ii}x_i + \sum_{j=i+1}^n u_{ij}x_j = b_i.$$

Starting at $i = n$, division by $u_{ii} \neq 0$ determines exactly one x_i at each step. The lower-triangular recursion follows from

$$\sum_{j=1}^{i-1} \ell_{ij} x_j + \ell_{ii} x_i = b_i$$

in the order $i = 1, \dots, n$. Replacing scalars by blocks gives the two block formulas; nonsingularity of each diagonal block gives existence and uniqueness at each step. $\square$

Example 5.4 (Two triangular solves).

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}}_{\mathbf{L}} \mathbf{y} = \begin{pmatrix} -8 \\ -23 \\ -7 \end{pmatrix} \implies \mathbf{y} = \begin{pmatrix} -8 \\ -7 \\ 6 \end{pmatrix},$$

and

$$\underbrace{\begin{pmatrix} 2 & 2 & -4 \\ 0 & -1 & -4 \\ 0 & 0 & 3 \end{pmatrix}}_{\mathbf{U}} \mathbf{x} = \mathbf{y} \implies \mathbf{x} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Thus $\mathbf{LUx} = (-8, -23, -7)'$, using two recursive solves and no matrix inverse.

Origin: Course formulation from H Eqs. 11.8.3 and 11.8.6

3.2 LDU and LU factorisations

Definition 5.17 (LDU and LU factorisations). An **LDU factorisation** of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is

$$\mathbf{A} = \mathbf{LDU},$$

where $\mathbf{L}$ is unit lower triangular, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is unit upper triangular; **unit** means that every diagonal entry is 1. An **LU factorisation** is $\mathbf{A} = \tilde{\mathbf{L}}\tilde{\mathbf{U}}$, where $\tilde{\mathbf{L}}$ is unit lower triangular and $\tilde{\mathbf{U}}$ is upper triangular. Every LDU factorisation gives the LU factorisation $\mathbf{A} = \mathbf{L}(\mathbf{DU})$.

Origin: H §14.5 preceding Lemma 14.5.1; S §11.1

Lemma 5.18 (Bordered LDU identities). For $n \geq 2$, partition

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_* & \mathbf{a} \\ \mathbf{b}' & c \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} \mathbf{L}_* & \mathbf{0} \\ \boldsymbol{\ell}' & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \mathbf{D}_* & \mathbf{0} \\ \mathbf{0}' & d \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{U}_* & \mathbf{u} \\ \mathbf{0}' & 1 \end{pmatrix}.$$

Then $\mathbf{A} = \mathbf{LDU}$ if and only if

$$\mathbf{A}_* = \mathbf{L}_* \mathbf{D}_* \mathbf{U}_*, \quad \mathbf{a} = \mathbf{L}_* \mathbf{D}_* \mathbf{u}, \quad \mathbf{b} = \mathbf{U}_*' \mathbf{D}_* \boldsymbol{\ell}, \quad d = c - \boldsymbol{\ell}' \mathbf{D}_* \mathbf{u}.$$

Origin: H Lemma 14.5.1

Proof. Block multiplication gives

$$\mathbf{LDU} = \begin{pmatrix} \mathbf{L}_* \mathbf{D}_* \mathbf{U}_* & \mathbf{L}_* \mathbf{D}_* \mathbf{u} \\ \boldsymbol{\ell}' \mathbf{D}_* \mathbf{U}_* & \boldsymbol{\ell}' \mathbf{D}_* \mathbf{u} + d \end{pmatrix}.$$

Equating the four blocks is equivalent to the displayed identities. □

Theorem 5.19 (Existence and recursive construction of LDU). For $i = 1, \dots, n$, let $\mathbf{A}_i$ be the leading $i \times i$ principal submatrix of $\mathbf{A} = (a_{ij})$, formed from rows and columns $1, \dots, i$. For $i \geq 2$, write

$$\mathbf{A}_i = \begin{pmatrix} \mathbf{A}_{i-1} & \mathbf{a}_i \\ \mathbf{b}'_i & a_{ii} \end{pmatrix}.$$

Then $\mathbf{A}$ has an LDU factorisation if and only if

$$\mathbf{a}_i \in \text{col}(\mathbf{A}_{i-1}), \quad \mathbf{b}'_i \in \text{row}(\mathbf{A}_{i-1}) \quad (i = 2, \dots, n).$$

Starting with $\mathbf{L}_1 = \mathbf{U}_1 = (1)$, $\mathbf{D}_1 = (a_{11})$, choose $\ell_i, \mathbf{u}_i$ satisfying

$$\mathbf{U}'_{i-1} \mathbf{D}_{i-1} \ell_i = \mathbf{b}_i, \quad \mathbf{L}_{i-1} \mathbf{D}_{i-1} \mathbf{u}_i = \mathbf{a}_i,$$

and set

$$d_i = a_{ii} - \ell'_i \mathbf{D}_{i-1} \mathbf{u}_i,$$

$$\mathbf{L}_i = \begin{pmatrix} \mathbf{L}_{i-1} & \mathbf{0} \\ \ell'_i & 1 \end{pmatrix}, \quad \mathbf{D}_i = \begin{pmatrix} \mathbf{D}_{i-1} & \mathbf{0} \\ \mathbf{0}' & d_i \end{pmatrix}, \quad \mathbf{U}_i = \begin{pmatrix} \mathbf{U}_{i-1} & \mathbf{u}_i \\ \mathbf{0}' & 1 \end{pmatrix}.$$

Then $\mathbf{A}_i = \mathbf{L}_i \mathbf{D}_i \mathbf{U}_i$ for every i .

Origin: H Thm. 14.5.4

Proof. Suppose the containment conditions hold and $\mathbf{A}_{i-1} = \mathbf{L}_{i-1} \mathbf{D}_{i-1} \mathbf{U}_{i-1}$. Write

$$\mathbf{a}_i = \mathbf{A}_{i-1} \mathbf{r}, \quad \mathbf{b}'_i = \mathbf{s}' \mathbf{A}_{i-1}.$$

Then

$$\mathbf{u}_i = \mathbf{U}_{i-1} \mathbf{r}, \quad \boldsymbol{\ell}_i = \mathbf{L}'_{i-1} \mathbf{s}$$

satisfy the two systems. Lemma 5.18 gives $\mathbf{A}_i = \mathbf{L}_i \mathbf{D}_i \mathbf{U}_i$; induction gives $i = 2, \dots, n$.

Conversely, an LDU factorisation of $\mathbf{A}$ restricts, by block multiplication, to an LDU factorisation of every $\mathbf{A}_i$. Lemma 5.18 gives

$$\mathbf{a}_i = \mathbf{L}_{i-1} \mathbf{D}_{i-1} \mathbf{u}_i = \mathbf{A}_{i-1} \mathbf{U}_{i-1}^{-1} \mathbf{u}_i \in \text{col}(\mathbf{A}_{i-1}),$$

and

$$\mathbf{b}'_i = \boldsymbol{\ell}'_i \mathbf{D}_{i-1} \mathbf{U}_{i-1} = \boldsymbol{\ell}'_i \mathbf{L}_{i-1}^{-1} \mathbf{A}_{i-1} \in \text{row}(\mathbf{A}_{i-1}).$$

□

Corollary 5.20 (Leading principal minors and uniqueness). Let $\Delta_i = \det(\mathbf{A}_i)$ and $\Delta_0 = 1$. If

$$\Delta_1 \Delta_2 \cdots \Delta_{n-1} \neq 0,$$

then $\mathbf{A}$ has a unique LDU factorisation. Its diagonal entries satisfy

$$d_i = \frac{\Delta_i}{\Delta_{i-1}} \quad (i = 1, \dots, n), \quad \det(\mathbf{A}) = \prod_{i=1}^n d_i.$$

If $\mathbf{A}$ is nonsingular, the condition is also necessary for an unpivoted LU factorisation with unit lower factor.

Origin: H Cor. 14.5.7 and Ex. 14.15; S Thm. 11.1

Proof. For $n = 1$, $\mathbf{A} = (1)(a_{11})(1)$, and all assertions hold. Assume $n \geq 2$. For $i \geq 2$, nonsingularity of $\mathbf{A}_{i-1}$ gives both containments in Theorem 5.19, hence existence. Uniqueness follows by induction. For the bordered step, the LDU factorisation of $\mathbf{A}_{i-1}$ is unique, and the nonsingular matrices $\mathbf{L}_{i-1}\mathbf{D}_{i-1}$ and $\mathbf{U}'_{i-1}\mathbf{D}_{i-1}$ determine $\mathbf{u}_i, \ell_i$ uniquely; then d_i is unique.

Triangular determinants give

$$\Delta_i = \det(\mathbf{L}_i) \det(\mathbf{D}_i) \det(\mathbf{U}_i) = \prod_{k=1}^i d_k,$$

so $d_i = \Delta_i / \Delta_{i-1}$ and $\det(\mathbf{A}) = \prod_i d_i$. If a nonsingular $\mathbf{A} = \mathbf{L}\tilde{\mathbf{U}}$, then its leading blocks satisfy $\mathbf{A}_i = \mathbf{L}_i\tilde{\mathbf{U}}_i$, and both triangular factors are nonsingular; hence $\Delta_i \neq 0$. $\square$

Theorem 5.21 (Scalar LDU recursion). Under the hypotheses of Corollary 5.20, the entries of $\mathbf{L} = (\ell_{ij})$, $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$, and $\mathbf{U} = (u_{ij})$ are generated, for $i = 1, \dots, n$, by

$$\begin{aligned} d_i &= a_{ii} - \sum_{k=1}^{i-1} \ell_{ik} d_k u_{ki}, \\ u_{ij} &= \frac{a_{ij} - \sum_{k=1}^{i-1} \ell_{ik} d_k u_{kj}}{d_i} & (j = i+1, \dots, n), \\ \ell_{ji} &= \frac{a_{ji} - \sum_{k=1}^{i-1} \ell_{jk} d_k u_{ki}}{d_i} & (j = i+1, \dots, n), \end{aligned}$$

where empty sums equal 0, $\ell_{ii} = u_{ii} = 1$, and the two divisions are performed only for $i < n$.

Origin: H §14.5e, Eq. 14.5.15

Proof. For $i \leq j$, the (i, j) entry of $\mathbf{LDU}$ is

$$a_{ij} = \sum_{k=1}^i \ell_{ik} d_k u_{kj} = \sum_{k=1}^{i-1} \ell_{ik} d_k u_{kj} + d_i u_{ij}.$$

Taking $j = i$ gives the first recursion; taking $j > i$ gives the second. For $j > i$, the (j, i) entry is

$$a_{ji} = \sum_{k=1}^i \ell_{jk} d_k u_{ki} = \sum_{k=1}^{i-1} \ell_{jk} d_k u_{ki} + \ell_{ji} d_i,$$

which gives the third. Corollary 5.20 gives $d_i \neq 0$ for $i < n$. □

Example 5.5 (An exact LDU factorisation).

$$\begin{pmatrix} 2 & 2 & -4 \\ 4 & 3 & -12 \\ -2 & -5 & -5 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}}_{\mathbf{L}} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}}_{\mathbf{D}} \underbrace{\begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}}_{\mathbf{U}}.$$

The leading principal minors are 2, -2 , -6 , and

$$(d_1, d_2, d_3) = \left(2, \frac{-2}{2}, \frac{-6}{-2}\right) = (2, -1, 3).$$

3.3 Row pivoting and repeated right-hand sides

Definition 5.22 (Exact pivoting and partial pivoting). After the first $k - 1$ pivot rows and columns have been processed, rows and columns $k, \dots, n$ form the **active submatrix**; its first diagonal entry is the **candidate pivot**. If that entry is zero while another entry in the active pivot column is nonzero, elimination in that column requires a row interchange; if the active pivot column is zero, no row interchange supplies a pivot there. **Partial pivoting** interchanges the candidate row with an active row whose pivot-column entry p has largest absolute value. The subsequent multipliers in that column satisfy

$$m_i = \frac{a_{ik}}{p}, \quad |m_i| \leq 1.$$

Origin: Course formulation from A&M Ex. 6.13; S §12 opening

Theorem 5.23 (Pivoted LU). For every nonsingular $\mathbf{A} \in \mathbb{R}^{n \times n}$, there exist a permutation matrix $\mathbf{P}$, a unit lower triangular matrix $\mathbf{L}$, and a nonsingular upper triangular matrix $\mathbf{U}$ such that

$$\mathbf{PA} = \mathbf{LU}.$$

Origin: Course formulation from A&M Ex. 6.13 and 6.26; S Thm. 11.1

Proof. Induct on n . For $n = 1$, take $\mathbf{P} = \mathbf{L} = (1)$, $\mathbf{U} = \mathbf{A}$. For $n > 1$, nonsingularity gives a nonzero entry in column 1. A row interchange $\mathbf{P}_1$ gives

$$\mathbf{P}_1\mathbf{A} = \begin{pmatrix} \alpha & \mathbf{r}' \\ \mathbf{c} & \mathbf{B} \end{pmatrix}, \quad \alpha \neq 0, \quad \mathbf{s} = \mathbf{c}/\alpha, \quad \mathbf{S} = \mathbf{B} - \mathbf{s}\mathbf{r}'.$$

The Schur determinant identity gives $\det(\mathbf{P}_1 \mathbf{A}) = \alpha \det(\mathbf{S})$, so $\mathbf{S}$ is nonsingular. By induction, $\mathbf{P}_2 \mathbf{S} = \mathbf{L}_2 \mathbf{U}_2$. Then

$$\underbrace{\begin{pmatrix} 1 & \mathbf{0}' \\ \mathbf{0} & \mathbf{P}_2 \end{pmatrix}}_{\mathbf{P}} \mathbf{P}_1 \mathbf{A} = \underbrace{\begin{pmatrix} 1 & \mathbf{0}' \\ \mathbf{P}_2 \mathbf{s} & \mathbf{L}_2 \end{pmatrix}}_{\mathbf{L}} \underbrace{\begin{pmatrix} \alpha & \mathbf{r}' \\ \mathbf{0} & \mathbf{U}_2 \end{pmatrix}}_{\mathbf{U}}.$$

The three displayed factors have the required forms. □

Corollary 5.24 (Factor once, solve many right-hand sides). If $\mathbf{PA} = \mathbf{LU}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, then

$$\mathbf{AX} = \mathbf{B} \iff \mathbf{LY} = \mathbf{PB}, \quad \mathbf{UX} = \mathbf{Y}.$$

The factors depend on $\mathbf{A}$, not on $\mathbf{B}$; each added column of $\mathbf{B}$ requires only one forward and one back substitution.

Origin: H §7.1 and Eqs. 11.8.2–11.8.6; S §11.1

Proof. Set $\mathbf{Y} = \mathbf{UX}$. Then $\mathbf{AX} = \mathbf{B} \iff \mathbf{PAX} = \mathbf{PB} \iff \mathbf{LY} = \mathbf{PB}$; nonsingularity of $\mathbf{L}, \mathbf{U}$ gives unique $\mathbf{Y}, \mathbf{X}$, columnwise. □

4 Computation: Numerical Checks in R

4.1 An elementary product and its RREF

```

1 A <- rbind(c(1, 2, -1, 0),
2             c(2, 4,  0, 2),
3             c(0, 0,  2, 2))
4 E1 <- diag(3); E1[2, 1] <- -2
5 E2 <- diag(3); E2[3, 2] <- -1
6 E3 <- diag(c(1, 1 / 2, 1))
7 E4 <- diag(3); E4[1, 2] <- 1
8 E <- E4 %*% E3 %*% E2 %*% E1
9 R_manual <- rbind(c(1, 2, 0, 1),
10                  c(0, 0, 1, 1),
11                  c(0, 0, 0, 0))
12 E %*% A - R_manual

```

```

##      [,1] [,2] [,3] [,4]
## [1,]    0    0    0    0
## [2,]    0    0    0    0
## [3,]    0    0    0    0

```

```
1 solve(E, R_manual) - A
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    0    0    0    0
## [2,]    0    0    0    0
## [3,]    0    0    0    0
```

4.2 Augmented tracking

```
1 A <- rbind(c(1, 2, -1), c(2, 4, 0))
2 b <- c(1, 4)
3 E1 <- rbind(c(1, 0), c(-2, 1))
4 E2 <- diag(c(1, 1 / 2))
5 E3 <- rbind(c(1, 1), c(0, 1))
6 E <- E3 %*% E2 %*% E1
7 aug_manual <- rbind(c(1, 2, 0, 2), c(0, 0, 1, 1))
8 E %*% cbind(A, b) - aug_manual
```

```
##      b
## [1,] 0 0 0 0
## [2,] 0 0 0 0
```

```
1 x0 <- c(2, 0, 1)
2 z <- c(-2, 1, 0)
3 cbind(particular_residual = A %*% x0 - b,
4        null_residual = A %*% z)
```

```
##      [,1] [,2]
## [1,]    0    0
## [2,]    0    0
```

4.3 LDU reconstruction, determinant, and solve

```
1 L <- rbind(c(1, 0, 0), c(2, 1, 0), c(-1, 3, 1))
2 D <- diag(c(2, -1, 3))
3 U <- rbind(c(1, 1, -2), c(0, 1, 4), c(0, 0, 1))
4 A <- rbind(c(2, 2, -4), c(4, 3, -12), c(-2, -5, -5))
5 b <- c(-8, -23, -7)
6 x_formula <- c(1, -1, 2)
7 x_package <- solve(A, b)
8 cbind(reconstruction = A - L %*% D %*% U,
9        solve_residual = A %*% x_formula - b)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    0    0    0    0
## [2,]    0    0    0    0
## [3,]    0    0    0    0
```

```
1 c(det_formula_difference = -6 - prod(diag(D)),
2   det_package_difference = det(A) - (-6),
3   solution_difference = max(abs(x_formula - x_package)))
```

```
## det_formula_difference det_package_difference
##                0.000e+00                -5.329e-15
##      solution_difference
##                5.773e-15
```

4.4 Pivoted LU with two right-hand sides

```
1 A <- rbind(c(0, 2, 1), c(1, 1, 0), c(2, 3, 4))
2 P <- rbind(c(0, 0, 1), c(0, 1, 0), c(1, 0, 0))
3 L <- rbind(c(1, 0, 0), c(1 / 2, 1, 0), c(0, -4, 1))
4 U <- rbind(c(2, 3, 4), c(0, -1 / 2, -2), c(0, 0, -7))
5 X_true <- cbind(c(1, -1, 2), c(0, 3, -2))
6 B <- A %*% X_true
```

```

7 Y <- solve(L, P %*% B)
8 X_factor <- solve(U, Y)
9 cbind(factor_residual = P %*% A - L %*% U,
10       solve_residual = A %*% X_factor - B)

```

```

##      [,1] [,2] [,3] [,4] [,5]
## [1,]    0    0    0    0    0
## [2,]    0    0    0    0    0
## [3,]    0    0    0    0    0

```

```

1 X_factor - solve(A, B)

```

```

##      [,1] [,2]
## [1,]    0    0
## [2,]    0    0
## [3,]    0    0

```

5 Pitfalls

5.1 A zero candidate pivot is not singularity

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det(\mathbf{A}) = -1, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The zero in position $(1, 1)$ prevents an unpivoted LDU factorisation because the first row and column are not both zero. One row interchange supplies a nonzero pivot.

5.2 Partial pivoting bounds column multipliers

$$\mathbf{A}_\varepsilon = \begin{pmatrix} \varepsilon & 1 \\ 1 & 1 \end{pmatrix}, \quad 0 < |\varepsilon| < 1.$$

No interchange is required in exact arithmetic. Without one, the elimination multiplier is $1/\varepsilon$; after interchanging the rows it is ε . Partial pivoting bounds these column multipliers; it makes no assertion about other arithmetic quantities. A small nonzero candidate pivot does not obstruct exact elimination.

5.3 Row operations do not preserve column space

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{E}_{12}\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

$$\text{col}(\mathbf{A}) = \text{span}\{(1, 0)'\} \neq \text{span}\{(0, 1)'\} = \text{col}(\mathbf{E}_{12}\mathbf{A}).$$

Row operations preserve row space, null space, and rank. Pivot indices in an RREF identify basis columns of the original matrix; the RREF columns themselves belong to a transformed column space.

5.4 The right-hand side must undergo the same operations

$$\mathbf{E}(\mathbf{A} : \mathbf{b}) = (\mathbf{EA} : \mathbf{Eb}).$$

Replacing this by $(\mathbf{EA} : \mathbf{b})$ changes the equations unless $\mathbf{Eb} = \mathbf{b}$. Eliminate the entire augmented matrix or retain $\mathbf{E}$.

6 Exercises

Exercise 5.1 (Rank-one formulas for elementary matrices). For $i \neq j$, derive

$$\mathbf{E}_{ij} = \mathbf{I} - (\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)', \quad \mathbf{E}_i(\lambda) = \mathbf{I} + (\lambda - 1)\mathbf{e}_i\mathbf{e}_i', \quad \mathbf{E}_i(\lambda \mid j) = \mathbf{I} + \lambda\mathbf{e}_i\mathbf{e}_j'.$$

Use the last formula to prove

$$\mathbf{E}_i(\lambda \mid j)\mathbf{E}_i(\mu \mid j) = \mathbf{E}_i(\lambda + \mu \mid j).$$

Origin: A&M Ex. 6.3 and 6.5

Solution. The identity decomposition $\mathbf{I} = \sum_k \mathbf{e}_k\mathbf{e}_k'$ gives

$$\mathbf{E}_{ij} = \sum_{k \notin \{i,j\}} \mathbf{e}_k\mathbf{e}_k' + \mathbf{e}_i\mathbf{e}_j' + \mathbf{e}_j\mathbf{e}_i' = \mathbf{I} - (\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)',$$

$$\mathbf{E}_i(\lambda) = \sum_{k \neq i} \mathbf{e}_k\mathbf{e}_k' + \lambda\mathbf{e}_i\mathbf{e}_i' = \mathbf{I} + (\lambda - 1)\mathbf{e}_i\mathbf{e}_i',$$

$$\mathbf{E}_i(\lambda \mid j) = \mathbf{I} + \lambda\mathbf{e}_i\mathbf{e}_j'.$$

Since $\mathbf{e}'_j \mathbf{e}_i = 0$,

$$(\mathbf{I} + \lambda \mathbf{e}_i \mathbf{e}'_j)(\mathbf{I} + \mu \mathbf{e}_i \mathbf{e}'_j) = \mathbf{I} + (\lambda + \mu) \mathbf{e}_i \mathbf{e}'_j.$$

Exercise 5.2 (Retaining the augmented transformation). For

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \end{pmatrix},$$

find $\mathbf{E}$ such that

$$\mathbf{E}(\mathbf{A} : \mathbf{b}) = \begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Use the transformed system to give every solution.

Origin: Course formulation from A&M Ex. 6.21

Solution. Apply $R_2 \leftarrow R_2 - 2R_1$, $R_2 \leftarrow R_2/3$, and $R_1 \leftarrow R_1 + R_2$:

$$\mathbf{E} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & 1/3 \\ -2/3 & 1/3 \end{pmatrix}.$$

The transformed equations are $x_1 + 2x_2 = 2$, $x_3 = 1$. Hence

$$\mathbf{x} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Exercise 5.3 (Coordinates for a nonhomogeneous solution set). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank r , let $\mathbf{B} \neq \mathbf{O}$, and suppose $\mathbf{AX} = \mathbf{B}$, $\mathbf{X} \in \mathbb{R}^{n \times p}$, has a solution $\mathbf{X}_0$. If $\mathbf{z}_1, \dots, \mathbf{z}_{n-r}$ is a basis of $\text{null}(\mathbf{A})$, set

$$\mathbf{Z}_{jk} = \mathbf{z}_j \mathbf{e}'_k, \quad \mathbf{X}_{jk} = \mathbf{X}_0 + \mathbf{Z}_{jk} \quad (1 \leq j \leq n-r, 1 \leq k \leq p).$$

Prove:

- (a) the matrices $\mathbf{Z}_{jk}$ form a basis of $\{\mathbf{Z} \in \mathbb{R}^{n \times p} : \mathbf{AZ} = \mathbf{O}\}$;
- (b) $\mathbf{X}_0$ and the $\mathbf{X}_{jk}$ are linearly independent;
- (c)

$$\{\mathbf{X} : \mathbf{AX} = \mathbf{B}\} = \left\{ c_0 \mathbf{X}_0 + \sum_{j,k} c_{jk} \mathbf{X}_{jk} : c_0 + \sum_{j,k} c_{jk} = 1 \right\};$$

- (d) the solution set is not a linear subspace of $\mathbb{R}^{n \times p}$.

Origin: Course formulation from H Ex. 11.4

Solution. For $\mathbf{Z} = (\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(p)})$,

$$\mathbf{AZ} = \mathbf{O} \iff \mathbf{z}^{(k)} = \sum_{j=1}^{n-r} a_{jk} \mathbf{z}_j \quad (1 \leq k \leq p) \iff \mathbf{Z} = \sum_{j,k} a_{jk} \mathbf{Z}_{jk}.$$

Columnwise comparison proves independence of the $\mathbf{Z}_{jk}$, hence (a).

If

$$c_0 \mathbf{X}_0 + \sum_{j,k} c_{jk} \mathbf{X}_{jk} = \mathbf{0}, \quad s = c_0 + \sum_{j,k} c_{jk},$$

then applying $\mathbf{A}$ gives $s\mathbf{B} = \mathbf{0}$, so $s = 0$. Thus $\sum_{j,k} c_{jk} \mathbf{Z}_{jk} = \mathbf{0}$, whence every $c_{jk} = 0$ and $c_0 = 0$. This proves (b).

When the coefficients sum to 1,

$$c_0 \mathbf{X}_0 + \sum_{j,k} c_{jk} \mathbf{X}_{jk} = \mathbf{X}_0 + \sum_{j,k} c_{jk} \mathbf{Z}_{jk},$$

which is a solution. Conversely, every solution is $\mathbf{X}_0 + \sum_{j,k} a_{jk} \mathbf{Z}_{jk}$; take $c_{jk} = a_{jk}$ and $c_0 = 1 - \sum_{j,k} a_{jk}$. This proves (c). Finally, $\mathbf{0}$ is not a solution because $\mathbf{B} \neq \mathbf{0}$, proving (d). ■

Exercise 5.4 (A discrete difference system). For $n \geq 2$, let

$$\mathbf{D} = (\mathbf{e}_1 - \mathbf{e}_2, \mathbf{e}_2 - \mathbf{e}_3, \dots, \mathbf{e}_{n-1} - \mathbf{e}_n) \in \mathbb{R}^{n \times (n-1)}.$$

Prove

$$\text{rank}(\mathbf{D}) = n - 1, \quad \text{null}(\mathbf{D}') = \text{span}\{\mathbf{1}_n\}.$$

Deduce that $\mathbf{D}\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{1}'_n \mathbf{b} = 0$, and in that case prove that its unique solution is

$$x_j = \sum_{i=1}^j b_i \quad (1 \leq j \leq n-1).$$

Origin: Course formulation from H Thm. 7.3.1

Solution. For $\mathbf{x} \in \mathbb{R}^{n-1}$, the first $n-1$ coordinates of $\mathbf{D}\mathbf{x} = \mathbf{0}$ give

$$x_1 = 0, \quad -x_{j-1} + x_j = 0 \quad (2 \leq j \leq n-1),$$

so $\mathbf{x} = \mathbf{0}$ and $\text{rank}(\mathbf{D}) = n-1$. Also

$$\mathbf{D}'\mathbf{k} = \mathbf{0} \iff k_i - k_{i+1} = 0 \quad (1 \leq i < n) \iff \mathbf{k} = c\mathbf{1}_n.$$

Consistency and compatibility therefore give

$$\mathbf{D}\mathbf{x} = \mathbf{b} \text{ is consistent} \iff \mathbf{k}'\mathbf{b} = 0 \quad (\mathbf{k} \in \text{null}(\mathbf{D}')) \iff \mathbf{1}'_n \mathbf{b} = 0.$$

Under this condition, the stated x_j satisfy

$$(\mathbf{D}\mathbf{x})_1 = b_1, \quad (\mathbf{D}\mathbf{x})_j = -x_{j-1} + x_j = b_j \quad (2 \leq j < n), \quad (\mathbf{D}\mathbf{x})_n = -x_{n-1} = b_n.$$

Full column rank gives uniqueness. ■

Exercise 5.5 (Two-sided annihilation at nullity one). Let $\mathbf{A}, \mathbf{Z} \in \mathbb{R}^{n \times n}$, $\text{rank}(\mathbf{A}) = n - 1$, and let nonzero $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ satisfy

$$\mathbf{A}\mathbf{x} = \mathbf{0}, \quad \mathbf{A}'\mathbf{y} = \mathbf{0}.$$

Prove

- (a) $\mathbf{AZ} = \mathbf{0}$ if and only if $\mathbf{Z} = \mathbf{x}\mathbf{k}'$ for some $\mathbf{k} \in \mathbb{R}^n$;
- (b) $\mathbf{AZ} = \mathbf{ZA} = \mathbf{0}$ if and only if $\mathbf{Z} = c\mathbf{x}\mathbf{y}'$ for some $c \in \mathbb{R}$.

Origin: H Ex. 11.3

Solution. Rank-nullity gives

$$\text{null}(\mathbf{A}) = \text{span}\{\mathbf{x}\}, \quad \text{null}(\mathbf{A}') = \text{span}\{\mathbf{y}\}.$$

- (a) $\mathbf{AZ} = \mathbf{0}$ exactly when every column of $\mathbf{Z}$ belongs to $\text{span}\{\mathbf{x}\}$, exactly when $\mathbf{Z} = \mathbf{x}\mathbf{k}'$.

(b) By (a), $\mathbf{Z} = \mathbf{x}\mathbf{k}'$. Since $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{Z}\mathbf{A} = \mathbf{0} \iff \mathbf{k}'\mathbf{A} = \mathbf{0}' \iff \mathbf{k} \in \text{null}(\mathbf{A}') \iff \mathbf{k} = c\mathbf{y}.$$

The reverse implications are the same equalities read backward. ■

Exercise 5.6 (Premultiplication and the complete solution set). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{m \times p}$, and $\mathbf{C} \in \mathbb{R}^{q \times m}$, where $p \geq 1$. Assume $\mathbf{C}\mathbf{A}\mathbf{X} = \mathbf{C}\mathbf{B}$ is consistent. Prove that it has exactly the same solutions as $\mathbf{A}\mathbf{X} = \mathbf{B}$ if and only if

$$\text{rank}(\mathbf{C}(\mathbf{A} : \mathbf{B})) = \text{rank}(\mathbf{A} : \mathbf{B}).$$

Origin: H Ex. 11.7

Solution. Set $\mathbf{M} = (\mathbf{A} : \mathbf{B})$. Since $\text{row}(\mathbf{C}\mathbf{M}) \subseteq \text{row}(\mathbf{M})$, rank equality gives a matrix $\mathbf{D}$ with $\mathbf{M} = \mathbf{D}\mathbf{C}\mathbf{M}$. Thus

$$\mathbf{C}\mathbf{A}\mathbf{X} = \mathbf{C}\mathbf{B} \implies \mathbf{C}\mathbf{M} \begin{pmatrix} \mathbf{X} \\ -\mathbf{I}_p \end{pmatrix} = \mathbf{0} \implies \mathbf{M} \begin{pmatrix} \mathbf{X} \\ -\mathbf{I}_p \end{pmatrix} = \mathbf{0},$$

so $\mathbf{AX} = \mathbf{B}$. The reverse inclusion of solution sets always holds.

Conversely, let $\mathbf{X}_0$ solve the premultiplied system. Equivalence gives $\mathbf{AX}_0 = \mathbf{B}$. For $\mathbf{z} \in \text{null}(\mathbf{CA})$, $\mathbf{X}_0 + \mathbf{ze}'_1$ also solves the premultiplied system; equivalence gives $\mathbf{Az} = \mathbf{0}$. Hence

$$\text{null}(\mathbf{CA}) = \text{null}(\mathbf{A}), \quad \text{rank}(\mathbf{CA}) = \text{rank}(\mathbf{A}).$$

Since $\mathbf{B} = \mathbf{AX}_0$ and $\mathbf{CB} = \mathbf{CAX}_0$,

$$\text{rank}(\mathbf{A} : \mathbf{B}) = \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{CA}) = \text{rank}(\mathbf{CA} : \mathbf{CB})$$

Exercise 5.7 (When $\mathbf{AX}$ is invariant across solutions). Let $\mathbf{A} \in \mathbb{R}^{q \times n}$, $\mathbf{C} \in \mathbb{R}^{m \times q}$, and $\mathbf{B} \in \mathbb{R}^{m \times p}$, where $p \geq 1$. Suppose $\mathbf{CAX} = \mathbf{B}$ is consistent. Prove that $\mathbf{AX}$ has the same value for every solution if and only if

$$\text{rank}(\mathbf{CA}) = \text{rank}(\mathbf{A}).$$

Origin: H Ex. 11.8

Solution. Since $\text{null}(\mathbf{A}) \subseteq \text{null}(\mathbf{CA})$, rank equality is equivalent to $\text{null}(\mathbf{A}) = \text{null}(\mathbf{CA})$. For two solutions,

$$\mathbf{CA}(\mathbf{X}_1 - \mathbf{X}_2) = \mathbf{0} \implies \mathbf{A}(\mathbf{X}_1 - \mathbf{X}_2) = \mathbf{0},$$

which proves invariance.

Conversely, fix a solution $\mathbf{X}_0$. For every $\mathbf{z} \in \text{null}(\mathbf{CA})$, $\mathbf{X}_0 + \mathbf{ze}'_1$ is another solution. Invariance gives $\mathbf{Aze}'_1 = \mathbf{0}$, hence $\mathbf{Az} = \mathbf{0}$. Therefore the two null spaces, and hence the two ranks, are equal.



Exercise 5.8 (A matrix requiring a row pivot). Let

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 4 \end{pmatrix}.$$

Prove that $\mathbf{A}$ has no unpivoted LDU factorisation. Find $\mathbf{P}, \mathbf{L}, \mathbf{U}$ such that $\mathbf{PA} = \mathbf{LU}$.

Origin: Course formulation from H Cor. 14.5.7; S Thm. 11.1

Solution. If $\mathbf{A} = \mathbf{LDU}$, its first bordered step requires

$$(2) \in \text{col}((0)) = \{0\},$$

a contradiction. Interchange rows 1, 3 and eliminate:

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & -4 & 1 \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} 2 & 3 & 4 \\ 0 & -1/2 & -2 \\ 0 & 0 & -7 \end{pmatrix}.$$

Direct multiplication gives

$$\mathbf{LU} = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} = \mathbf{PA}.$$



Exercise 5.9 (When an LDU factorisation is unique). Suppose

$$\mathbf{A} = \mathbf{L} \operatorname{diag}(d_1, \dots, d_n) \mathbf{U}$$

is an LDU factorisation. Prove that it is unique if and only if

$$d_i \neq 0 \quad (1 \leq i < n),$$

equivalently, if and only if every leading principal submatrix of order less than n is nonsingular.

Origin: H Ex. 14.16(a) and 14.19

Solution. For the leading principal submatrix $\mathbf{A}_k$,

$$\det(\mathbf{A}_k) = \prod_{j=1}^k d_j.$$

Thus $d_i \neq 0$ for $i < n$ is equivalent to $\det(\mathbf{A}_k) \neq 0$ for $k < n$, and Corollary 5.20 gives uniqueness.

Conversely, if $d_i = 0$ for some $i < n$, then for every $t \in \mathbb{R}$,

$$\mathbf{U}_t = \mathbf{U} + t\mathbf{e}_i\mathbf{e}'_{i+1}$$

is unit upper triangular and

$$\text{diag}(d_1, \dots, d_n)\mathbf{U}_t = \text{diag}(d_1, \dots, d_n)\mathbf{U}.$$

The choices $t = 0$ and $t = 1$ give distinct LDU factorisations of $\mathbf{A}$. ■

Exercise 5.10 (Determinants from LDU pivots). Let $\mathbf{A} = \mathbf{L} \text{diag}(d_1, \dots, d_n)\mathbf{U}$ be an LDU factorisation, and let Δ_i be the determinant of its leading $i \times i$ principal submatrix, with $\Delta_0 = 1$. Prove

$$\Delta_i = \prod_{k=1}^i d_k, \quad d_i = \frac{\Delta_i}{\Delta_{i-1}} \quad (\Delta_{i-1} \neq 0), \quad \det(\mathbf{A}) = \prod_{k=1}^n d_k.$$

Evaluate $\det(\mathbf{A})$ when $(d_1, d_2, d_3, d_4) = (2, -3, 1/2, 5)$.

Origin: H Ex. 14.15 and §14.5e; S Thm. 11.1

Solution. The leading factors satisfy $\mathbf{A}_i = \mathbf{L}_i \mathbf{D}_i \mathbf{U}_i$, so

$$\Delta_i = \det(\mathbf{L}_i) \det(\mathbf{D}_i) \det(\mathbf{U}_i) = 1 \left(\prod_{k=1}^i d_k \right) 1.$$

Division of consecutive identities gives the ratio. At $i = n$ it gives the determinant formula. In the stated case,

$$\det(\mathbf{A}) = 2(-3)(1/2)5 = -15.$$

Exercise 5.11 (Substitution cost for many right-hand sides). Let $\mathbf{PA} = \mathbf{LU}$, where $\mathbf{L}$ is unit lower triangular and $\mathbf{U}$ is nonsingular upper triangular. For $\mathbf{B} \in \mathbb{R}^{n \times p}$, count the multiplication-subtractions and divisions needed after the factors are available to solve $\mathbf{AX} = \mathbf{B}$. State the saving relative to refactorising for every column.

Origin: H §7.1 and Eqs. 11.8.2–11.8.6; S Prop. 11.1

Solution. For one right-hand side, unit-lower substitution uses

$$\sum_{i=1}^n (i-1) = \frac{n(n-1)}{2}$$

multiplication-subtractions and no divisions. Upper substitution uses the same number of multiplication-subtractions and n divisions. For p columns, the total is

$$pn(n-1) \text{ multiplication-subtractions,} \quad pn \text{ divisions.}$$

The factorisation is performed once; separate solution would repeat it p times, so factor-once avoids $p-1$ factorisations. 

Exercise 5.12 (Rank-revealing rectangular LDU). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank $r > 0$.

(a) Prove that permutation matrices $\mathbf{P}, \mathbf{Q}$ can be chosen so that, in

$$\mathbf{PAQ} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix},$$

the $r \times r$ block $\mathbf{B}_{11}$ and all its leading principal submatrices are nonsingular.

(b) Prove the unique factorisation

$$\mathbf{PAQ} = \begin{pmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{pmatrix} \mathbf{D} (\mathbf{U}_1 \ \mathbf{U}_2),$$

where $\mathbf{B}_{11} = \mathbf{L}_1 \mathbf{D} \mathbf{U}_1$ is its LDU factorisation, and derive

$$\mathbf{L}_2 = \mathbf{B}_{21} \mathbf{U}_1^{-1} \mathbf{D}^{-1}, \quad \mathbf{U}_2 = \mathbf{D}^{-1} \mathbf{L}_1^{-1} \mathbf{B}_{12}.$$

Origin: H Ex. 14.20

Solution. (a) Rank r gives a nonsingular $r \times r$ submatrix. Row and column permutations move it to the leading block $\mathbf{C}$. Pivoted LU gives $\mathbf{P}_0 \mathbf{C} = \mathbf{L}_0 \mathbf{U}_0$. Extend $\mathbf{P}_0$ by the identity on the remaining rows. The leading determinant of order $i \leq r$ is

$$\det((\mathbf{L}_0)_i) \det((\mathbf{U}_0)_i) = \prod_{k=1}^i u_{kk} \neq 0.$$

(b) Corollary 5.20 gives the unique $\mathbf{B}_{11} = \mathbf{L}_1 \mathbf{D} \mathbf{U}_1$, with all factors nonsingular. Define $\mathbf{L}_2, \mathbf{U}_2$ by the displayed formulas. Since $\text{rank}(\mathbf{P} \mathbf{A} \mathbf{Q}) = \text{rank}(\mathbf{B}_{11}) = r$, the Schur rank identity gives

$$\mathbf{B}_{22} - \mathbf{B}_{21} \mathbf{B}_{11}^{-1} \mathbf{B}_{12} = \mathbf{O}.$$

Therefore

$$\begin{pmatrix} \mathbf{L}_1 \\ \mathbf{L}_2 \end{pmatrix} \mathbf{D} \begin{pmatrix} \mathbf{U}_1 & \mathbf{U}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{21} \mathbf{B}_{11}^{-1} \mathbf{B}_{12} \end{pmatrix} = \mathbf{P} \mathbf{A} \mathbf{Q}.$$

The leading block fixes $\mathbf{L}_1, \mathbf{D}, \mathbf{U}_1$; the two off-diagonal block equalities then fix $\mathbf{L}_2, \mathbf{U}_2$. ■

Exercise 5.13 (A block LU solve through a Schur complement). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \quad \mathbf{S} = \mathbf{D} - \mathbf{C} \mathbf{A}^{-1} \mathbf{B},$$

where $\mathbf{A}$ and $\mathbf{S}$ are nonsingular. Derive a solution of

$$\mathbf{M} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\eta} \\ \mathbf{g} \end{pmatrix}$$

using one solve with $\mathbf{S}$ and two solves with $\mathbf{A}$. Do not form $\mathbf{M}^{-1}$.

Origin: H Lemma 14.5.1; S §11.1.1

Solution. Block elimination gives

$$\begin{pmatrix} \mathbf{I} & \mathbf{O} \\ -\mathbf{CA}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{S} \end{pmatrix}.$$

Thus compute

$$\mathbf{Az} = \boldsymbol{\eta}, \quad \mathbf{Sy} = \mathbf{g} - \mathbf{Cz}, \quad \mathbf{Ax} = \boldsymbol{\eta} - \mathbf{By}.$$

Then $\mathbf{Ax} + \mathbf{By} = \boldsymbol{\eta}$, and

$$\mathbf{Cx} + \mathbf{Dy} = \mathbf{CA}^{-1}(\boldsymbol{\eta} - \mathbf{By}) + \mathbf{Dy} = \mathbf{Cz} + \mathbf{Sy} = \mathbf{g}.$$

